SCIENCE INVESTIGATIONS
PLANNING AND REPORT WORKSHEET

NAME: SOLUTIONS - WAX PHASE CHANGE **CLASS:** ____________

OTHER GROUP MEMBERS: ______________________________

Aim • What are you going to investigate?

Determine the melting point (freezing point) of candle wax. (1)

Hypothesis • What do you think will happen? Explain why.

Equipment • List the equipment you will need.

Candle wax
Large test tube
Thermometer
250mL beaker
Bunsen burner
Tripod
Gauze mat
Retort stand
Boss head + clamp.

(1)

Method • Which variable(s) are you going to

change? (independent)

time (1)

measure? (dependent)

temperature (1)

keep the same? (controlled)

volume of wax. (2) (Any 2).
same thermometer
same test tube
same room conditions

• How will you make it a fair test?

- List the steps you will take to do the investigation and collect the results. Use labelled diagrams if appropriate.

Suitable diagram. (1)
Steps in logical sequence. (2)
2 trials. (1)

Results

- Record your results or observations. Use a table if appropriate.

2 tables required.
Headings + units (1)
Clearly set out. (1)

- Can your results be represented in a graph?

Line graph. (1)
Suitable scales (1)
Labels + units (1)
Plotting (1)

Discussion + Critical Analysis

- What do your results tell you? Are their any patterns or trends in your results?

Adequately describes the shape - shows a "plateau". (1).

- Using some Science ideas, can you explain why you got these patterns or trends?

  - Phase change involves no temperature change. (1)
  - Heat is removed and allows the particles to move closer together. (1)

- What things went wrong or affected the results you obtained? Suggest improvements for this investigation if possible.

Conclusion

- What is the relationship you found from this investigation? It should answer the question asked in the "aim".

Should state the average melting point. (1)